

Written exam QM 1 track B - 02/09/2025

The candidate must choose **three exercises** out of the four presented in the following. **Do not** submit a text with more than three exercises: only the first three ones, in the order assigned by the candidate, will be taken into account. On each sheet, please indicate **first and last name**, and the university registration number.

Exercise 1

Let's consider a spin 1/2 particle:

- a) Assuming that the state is such that the probabilities of having the spin oriented parallel or antiparallel to the z -axis are equal, determine the state of the system that maximizes the expectation value of S_x ;
- b) Assuming now that the state is described by a generic superposition of eigenstates of S_z :

$$|\psi\rangle = a|+\rangle + b|-\rangle, \quad (1)$$

determine the two constants a, b such that $\langle S_z \rangle = 0$, $\langle S_y \rangle = \hbar/2$.

Exercise 2

Consider an electron in a Hydrogen atom. Let the state of the system be described by an eigenstate of the Hamiltonian operator with quantum numbers n and l .

- a) Taking advantage of the "Useful Formulas" section below, prove that the dependence on the variable r of the radial wave functions $R_{nl}(r)$ with $l = n - 1$ is such that

$$R_{n(n-1)}(r) = \left(\frac{2}{na}\right)^n \sqrt{\frac{2}{na(2n)!}} r^{n-1} e^{-r/na}. \quad (2)$$

- b) Determine the variance of the distance from the nucleus r of an electron with quantum numbers $n = 2$, $l = 1$.
- c) Calculate the expectation values of $L_x^2 + L_y^2$ on the Hamiltonian eigenstates with $n = 2$.

Exercise 3

Consider a two-state physical system and the basis composed of the states $|\psi_1\rangle$ and $|\psi_2\rangle$, which are eigenstates of the Hamiltonian H_0 corresponding to the eigenvalues E_1, E_2 respectively. Now consider a new system with Hamiltonian $H_0 + W$, where W is a coupling term between the two levels. The matrix representation of W in the basis $\{|\psi_1\rangle, |\psi_2\rangle\}$ is:

$$W_{11} = W_{22} = 0, \quad (3)$$

$$W_{12} = W_{21} = \omega, \quad (4)$$

where ω is a real positive constant.

- Determine how the eigenstates of the Hamiltonian and the energy levels change with the introduction of the coupling term.
- If at time $t = 0$ the new system is certainly in state $|\psi_1\rangle$, at what time instants (if any) will the system still be in the same state?
- Compute the probability of finding the system in state $|\psi_2\rangle$ for $t > 0$.

Exercise 4

Two identical fermions of mass m , constrained to move along a segment of length L along the x axis, have a spin component along the z axis equal to $+\hbar/2$.

- What is the minimum system energy and the corresponding eigenfunction?
- If there is an interaction potential $H_I = k\delta(x_1 - x_2)$, where k is a small number, how does the value of the energy change using first-order perturbation theory?
- How would the result in *b*) change if the two fermions occupy a spin-singlet state?

Useful relations

Eigenfunctions of the Hydrogen atom hamiltonian:

$$\psi_{nlm}(r, \theta, \phi) = R_{nl}(r) Y_l^m(\theta, \phi), \quad (5)$$

$$R_{nl}(r) = \sqrt{\left(\frac{2}{na}\right)^3 \frac{(n-l-1)!}{2n(n+l)!}} e^{-r/na} \left(\frac{2r}{na}\right)^l L_{n-l-1}^{2l+1}(2r/na), \quad (6)$$

$$L_q^p(x) = (-)^p \left(\frac{d}{dx}\right)^p L_{p+q}(x), \quad (7)$$

$$L_q(x) = \frac{e^x}{q!} \left(\frac{d}{dx}\right)^q (e^{-x} x^q). \quad (8)$$

Eigenfunctions and eigenvalues of the hamiltonian for a 1D infinite square well ($0 < x < a$):

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a} x\right), \quad E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2}. \quad (9)$$

[ESERCIZIO 1] - SPIN $1/2$

a) $|+\rangle$ autostato di S_z con "spin up"
 $|-\rangle$ "spin down"

$$|\psi\rangle = a|+\rangle + b|-\rangle$$

a, b tali che $|a|^2 = |b|^2$ (probabilità spin
 parallelo all'asse z
 = prob. spin antiparallelo all'asse z)

normalizzazione: $|a|^2 = |b|^2 = \frac{1}{2}$

$\rightarrow$ posso scegliere $a = \frac{1}{\sqrt{2}}$, $b = \frac{1}{\sqrt{2}} e^{i\varphi}$,

l'importante è che abbiano una fase RELATIVA.

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|+\rangle + e^{i\varphi} |-\rangle)$$

$\langle S_x \rangle = \langle \psi | \hat{S}_x | \psi \rangle = \rightarrow$ base di autostati di $\hat{S}_z$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & e^{-i\varphi} \end{pmatrix} \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ e^{i\varphi} \end{pmatrix} =$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & e^{-i\vartheta} \end{pmatrix} \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ e^{i\vartheta} \end{pmatrix} =$$

$$= \frac{\hbar}{4} \begin{pmatrix} 1 & e^{-i\vartheta} \end{pmatrix} \begin{pmatrix} e^{i\vartheta} \\ 1 \end{pmatrix} = \frac{\hbar}{4} (e^{i\vartheta} + e^{-i\vartheta}) =$$

$$= \frac{\hbar}{4} 2 \cos \vartheta = \frac{\hbar}{2} \cos \vartheta$$

$$\max \langle S_x \rangle = \frac{\hbar}{2} \rightarrow \cos \vartheta = +1$$

$$\vartheta = 2\pi \cdot n, \quad n=0,1,2,\dots$$

$$e^{i\vartheta} = 1$$

$$|\psi\rangle_{\max S_x} = \frac{1}{\sqrt{2}} (|+\rangle + |-\rangle)$$

che è proprio autostato di S_x con autovalore massimo $(\hbar/2)$ ✓

$$b) \quad |4\rangle = a|+\rangle + b|-\rangle \quad \swarrow \text{vs base } \hat{S}_2$$

$$0 = \langle S_z \rangle = (a^* \ b^*) \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} =$$

$$= \frac{\hbar}{2} (a^* \ b^*) \begin{pmatrix} a \\ -b \end{pmatrix} = \frac{\hbar}{2} (|a|^2 - |b|^2)$$

$$\Rightarrow |a|^2 = |b|^2 \quad (1)$$

$$\frac{\hbar}{2} = \langle S_y \rangle = (a^* \ b^*) \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} =$$

$$= \frac{\hbar}{2} (a^* \ b^*) \begin{pmatrix} -ib \\ ia \end{pmatrix} = \frac{\hbar}{2} (-ia^*b + ia b^*) =$$

$$= -i \frac{\hbar}{2} (a^*b - a b^*) = -i \frac{\hbar}{2} 2i \operatorname{Im}(a^*b) =$$

$$\quad \downarrow$$

$$x - x^* = 2i \operatorname{Im} x \quad = \hbar \operatorname{Im}(a^*b)$$

$$\Rightarrow \frac{1}{2} = \operatorname{Im}(a^*b) \quad (2)$$

$$(1) \Rightarrow |a|^2 = |b|^2 = 1/2, \quad a = 1/\sqrt{2}$$

$$b = 1/\sqrt{2} e^{i\varphi}$$

1 fase relativa

$$\frac{1}{2} = \text{Im} (a^* b) \quad (2)$$

$$(2) + (1) : \quad \text{Im} \left(\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} e^{i\varphi} \right) =$$

$$= \frac{1}{2} \text{Im} (\cos \varphi + i \sin \varphi) = \frac{1}{2} \sin \varphi$$

$$\frac{1}{2} = \frac{1}{2} \sin \varphi \quad \Rightarrow \quad \varphi = \frac{\pi}{2}$$

$$\Rightarrow |\psi\rangle = \frac{1}{\sqrt{2}} \left(|+\rangle + e^{i\pi/2} |-\rangle \right) =$$

$$= \frac{1}{\sqrt{2}} \left(|+\rangle + i |-\rangle \right) \quad \checkmark$$

[ESERCIZIO 2] - ATOMO DI IDROGENO

$$L_{n-l-1}^{2l+1}(x), \quad l = n-1 \quad a)$$

$$2l+1 = 2n-2+1 = 2n-1$$

$$n-l-1 = n-n+1-1 = 0$$

$$L_0^{2n-1}(x) = (-1)^{2n-1} \left(\frac{d}{dx} \right)^{2n-1} L_{2n-1}^{2n-1}(x) =$$

(questo ci fa già capire che $L_0^{2n-1}(x)$ dovrebbe essere una costante)

$$= (-1)^{2n-1} \left(\frac{d}{dx} \right)^{2n-1} \left\{ \frac{e^x}{(2n-1)!} \left(\frac{d}{dx} \right)^{2n-1} (e^{-x} x^{2n-1}) \right\} =$$

$$= (-1)^{2n-1} \left(\frac{d}{dx} \right)^{2n-1} \left\{ \frac{e^x}{(2n-1)!} \left[(-1)^{2n-1} e^{-x} x^{2n-1} + \right. \right.$$

$$\left. \left. + e^{-x} (2n-1)! \right] \right\} =$$

$$= (-1)^{2n-1} \left(\frac{d}{dx} \right)^{2n-1} \left\{ \frac{(-1)^{2n-1}}{(2n-1)!} x^{2n-1} + 1 \right\} =$$

$$= (-1)^{2n-1} (-1)^{2n-1} \frac{(2n-1)!}{(2n-1)!} = [(-1)^{2n-1}]^2 = 1.$$

$$R_{n(n-1)}(z) = \left(\frac{2}{ha}\right)^{3/2} \left[\frac{0!}{\frac{2n(2n-1)!}{(2n)!}} \right]^{1/2} e^{-z/ha} \cdot \left(\frac{2z}{ha}\right)^{h-1} =$$

$$= \left(\frac{2}{ha}\right)^{3/2} \left(\frac{2}{ha}\right)^{n-1} z^{n-1} e^{-z/ha} \frac{1}{\sqrt{(2n)!}} =$$

$$= \left(\frac{2}{ha}\right)^n \sqrt{\frac{2}{ha} \cdot \frac{1}{(2n)!}} z^{n-1} e^{-z/ha} \quad \checkmark$$

b) $\langle r^2 \rangle = \langle r \rangle^2$ per ψ_{21m} , irrilevante

$$\langle r^2 \rangle = \int_0^{+\infty} dr r^2 \cdot r^2 |R_{21}(r)|^2$$

$$R_{21}(r) = \left(\frac{2}{2a}\right)^2 \sqrt{\frac{2}{2a} \cdot \frac{1}{4!}} e^{-r/2a} r =$$

$\underbrace{4!}_{4 \cdot 3 \cdot 2 \cdot 1 = 24}$

$$= \frac{1}{a^2} \sqrt{\frac{1}{a} \cdot \frac{1}{24}} r e^{-r/2a}$$

$$|R_{21}(r)|^2 = \frac{1}{a^5} \frac{1}{24} r^2 e^{-r/a}$$

$$\langle r^2 \rangle = \frac{1}{24 a^5} \int_0^{+\infty} dr r^6 e^{-r/a} =$$

$$= \frac{1}{24 a^5} 6! a^6 \int_0^{+\infty} dr e^{-r/a} =$$

$$= \frac{1}{24 a^5} 6! a^6 (-a) \left[e^{-r/a} \right]_0^{+\infty} =$$

$$= \frac{6! a^6 a}{24 a^5} = \frac{6 \cdot 5 \cdot \cancel{4!}}{\cancel{4!}} a^2 = 30 a^2$$

per parti o derivando rispetto a $1/a$
(visto nelle esercitazioni del capitolo 4)

$$\frac{d}{dr} e^{-r/a} = -\frac{1}{a} e^{-r/a}$$

$$\langle r \rangle = \frac{1}{24 a^5} \int_0^{+\infty} dr r^5 e^{-r/a} =$$

$$= \frac{1}{4! a^5} 5! a^5 \int_0^{+\infty} dr e^{-r/a} =$$

$$= \frac{5! a^5 \cdot a}{4! a^5} = 5 a$$

$$\langle r^2 \rangle - \langle r \rangle^2 \Big|_{\substack{n=2 \\ l=1}} = 30 a^2 - 25 a^2 = 5 a^2 \quad \checkmark$$

$$c) \quad n=2 \rightarrow l=0, m=0$$

$$l=1, m=1, 0, -1$$

$$L_x^2 + L_y^2 = L^2 - L_z^2$$

$$L^2 |\psi_{nlm}\rangle = \hbar^2 l(l+1) |\psi_{nlm}\rangle$$

$$L_z |\psi_{nlm}\rangle = \hbar m |\psi_{nlm}\rangle$$

$$\langle \psi_{200} | L^2 - L_z^2 | \psi_{200} \rangle = 0$$

$$\langle \psi_{210} | L^2 - L_z^2 | \psi_{210} \rangle = \hbar^2 2$$

$$\langle \psi_{211} | L^2 - L_z^2 | \psi_{211} \rangle = 2\hbar^2 - \hbar^2 = \hbar^2$$

$$\langle \psi_{21-1} | L^2 - L_z^2 | \psi_{21-1} \rangle = 2\hbar^2 - (-\hbar)^2 = \hbar^2$$

{ ESERCIZIO 3 } - SISTEMA A DUE STATI

$$\begin{cases} \hat{H}_0 |\psi_1\rangle = E_1 |\psi_1\rangle \\ \hat{H}_0 |\psi_2\rangle = E_2 |\psi_2\rangle \\ \langle \psi_i | \psi_j \rangle = \delta_{ij} \end{cases}$$

$$\langle \psi_i | W | \psi_j \rangle = \begin{pmatrix} 0 & w \\ w & 0 \end{pmatrix}, \quad w \in \mathbb{R}^+$$

$i, j = 1, 2$

$$H' = H_0 + W$$

$$\langle \psi_i | H' | \psi_j \rangle = \begin{pmatrix} E_1 & w \\ w & E_2 \end{pmatrix}$$

$i, j = 1, 2$

a)

$$\begin{aligned} 0 &= \det \left(\langle \psi_i | H' | \psi_j \rangle - \lambda \mathbb{1} \right) = \\ &= \det \begin{pmatrix} E_1 - \lambda & w \\ w & E_2 - \lambda \end{pmatrix} = (E_1 - \lambda)(E_2 - \lambda) - w^2 = \\ &= E_1 E_2 - \lambda E_1 - \lambda E_2 + \lambda^2 - w^2 \end{aligned}$$

$$E_1 E_2 - \lambda E_1 - \lambda E_2 + \lambda^2 - \omega^2 = 0$$

$$\lambda^2 - \lambda(E_1 + E_2) + (E_1 E_2 - \omega^2) = 0$$

$$\lambda_{1,2} = \frac{(E_1 + E_2) \pm \sqrt{(E_1 + E_2)^2 - 4(E_1 E_2 - \omega^2)}}{2} \rightarrow \sqrt{\Delta}$$

$$\Delta = (E_1 + E_2)^2 - 4E_1 E_2 + 4\omega^2 =$$

$$= E_1^2 + E_2^2 - 2E_1 E_2 + 4\omega^2 = (E_1 - E_2)^2 + 4\omega^2 > 0$$

$$\lambda_1 = \frac{1}{2} \left[(E_1 + E_2) + \sqrt{(E_1 - E_2)^2 + 4\omega^2} \right] \doteq \frac{1}{2} (E_1 + E_2 + \epsilon)$$

$$\lambda_2 = \frac{1}{2} \left[(E_1 + E_2) - \sqrt{(E_1 - E_2)^2 + 4\omega^2} \right] \doteq \frac{1}{2} (E_1 + E_2 - \epsilon) \\ = \lambda_1 - \epsilon$$

tali che

$$\begin{cases} H' |\psi'_1\rangle = \lambda_1 |\psi'_1\rangle \\ H' |\psi'_2\rangle = \lambda_2 |\psi'_2\rangle \end{cases}$$

Scriviamo $|\psi'_{1,2}\rangle$ sulla base di $|\psi_{1,2}\rangle$:

$$\left\{ H' |\psi_1'\rangle = \lambda_1 |\psi_1'\rangle \right\}$$

$$\begin{pmatrix} E_1 & \omega \\ \omega & E_2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \lambda_1 \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\begin{cases} E_1 a + \omega b = \lambda_1 a \\ \omega a + E_2 b = \lambda_1 b \end{cases}$$

$$\omega b = (\lambda_1 - E_1) a, \quad b = \frac{\lambda_1 - E_1}{\omega} a$$

$$\omega a = \frac{(\lambda_1 - E_2)(\lambda_1 - E_1)}{\omega} a \rightarrow a \text{ parametro libero}$$

$$\left[\begin{aligned} \lambda_1 - E_1 &= \frac{1}{2}(E_1 + E_2 + \lambda) - E_1 = -\frac{1}{2}E_1 + \frac{1}{2}E_2 + \frac{\lambda}{2} = \\ &= \frac{1}{2}(-E_1 + E_2 + \lambda) \end{aligned} \right]$$

$$\lambda_1 - E_2 = \frac{1}{2}(E_1 - E_2 + \lambda)$$

$$(\lambda_1 - E_1)(\lambda_1 - E_2) = \frac{1}{4} \left\{ \left[\lambda + (E_1 - E_2) \right] \left[\lambda - (E_1 - E_2) \right] \right\} =$$

$$\left[\begin{aligned} &= \frac{1}{4} \left\{ \lambda^2 - (E_1 - E_2)^2 \right\} = \frac{1}{4} \left\{ (E_1 - E_2)^2 + 4\omega^2 - (E_1 - E_2)^2 \right\} = \\ &= \omega^2 \end{aligned} \right]$$

$$\Rightarrow \begin{pmatrix} \langle \psi_1 | \psi'_1 \rangle \\ \langle \psi_2 | \psi'_1 \rangle \end{pmatrix} = \psi'_1 = a \begin{pmatrix} 1 \\ \frac{\lambda_1 - E_1}{\omega} \end{pmatrix}$$

NORMALIZATION:

$$1 = |a|^2 \left(1 + \frac{(\lambda_1 - E_1)^2}{\omega^2} \right)$$

$$\omega^2 = |a|^2 (\omega^2 + (\lambda_1 - E_1)^2)$$

$$\psi'_1 = \frac{e^{i\varphi_a}}{\sqrt{\omega^2 + (\lambda_1 - E_1)^2}} \begin{pmatrix} \omega \\ \lambda_1 - E_1 \end{pmatrix}$$

— 0 —

$$\left\{ H' |\psi'_2\rangle = \lambda_2 |\psi'_2\rangle \right\}$$

$$\begin{pmatrix} E_1 & \omega \\ \omega & E_2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \lambda_2 \begin{pmatrix} a \\ b \end{pmatrix}, \quad \psi'_2 = \begin{pmatrix} \langle \psi_1 | \psi'_2 \rangle \\ \langle \psi_2 | \psi'_2 \rangle \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\begin{cases} E_1 a + \omega b = \lambda_2 a \\ \omega a + E_2 b = \lambda_2 b \end{cases}$$

$$\psi'_2 = \frac{e^{i\varphi'_a}}{\sqrt{\omega^2 + (\lambda_2 - E_1)^2}} \begin{pmatrix} \omega \\ \lambda_2 - E_1 \end{pmatrix}$$

$$b) |\psi'(t=0)\rangle \equiv |\psi_1\rangle$$

$$\begin{aligned}
 |\psi'(t)\rangle &= U(t,0)|\psi'(t=0)\rangle = e^{-\frac{i}{\hbar}H't} |\psi'(t=0)\rangle = \\
 &= e^{-\frac{i}{\hbar}\lambda_1 t} |\psi_1'\rangle \langle\psi_1'|\psi_1\rangle + e^{-\frac{i}{\hbar}\lambda_2 t} |\psi_2'\rangle \langle\psi_2'|\psi_1\rangle = \\
 &= e^{-\frac{i}{\hbar}\lambda_1 t} |\psi_1'\rangle \frac{e^{-i\varphi_2} \omega}{\sqrt{\omega^2 + (\lambda_1 - E_1)^2}} + \\
 &\quad + e^{-\frac{i}{\hbar}\lambda_2 t} |\psi_2'\rangle \frac{e^{-i\varphi_2'} \omega}{\sqrt{\omega^2 + (\lambda_2 - E_1)^2}} = \\
 &= e^{-\frac{i}{\hbar}\lambda_1 t} \frac{\omega}{\omega^2 + (\lambda_1 - E_1)^2} \begin{pmatrix} \omega \\ \lambda_1 - E_1 \end{pmatrix} + \\
 &\quad + e^{-\frac{i}{\hbar}\lambda_2 t} \frac{\omega}{\omega^2 + (\lambda_2 - E_1)^2} \begin{pmatrix} \omega \\ \lambda_2 - E_1 \end{pmatrix}.
 \end{aligned}$$

Cerchiamo t_1 t.c. $P(|\psi'(t_1)\rangle \equiv |\psi_1\rangle) = 1$

$$1 = |\langle\psi_1|\psi'(t_1)\rangle|^2.$$

Let's look at the projection of $|\psi'(t)\rangle$

along $|\psi_2\rangle$:

$$\langle \psi_2 | \psi'(t_1) \rangle =$$

$$= e^{-\frac{i}{\hbar} \lambda_1 t_1} \frac{\omega(\lambda_1 - E_1)}{\omega^2 + (\lambda_1 - E_1)^2} + \int |\psi_2\rangle \downarrow a|\psi_1'\rangle + b|\psi_2'\rangle$$

calcolare questo

$$+ e^{-\frac{i}{\hbar} \lambda_2 t_1} \frac{\omega(\lambda_2 - E_1)}{\omega^2 + (\lambda_2 - E_1)^2}$$

$$\left\{ \begin{array}{l} \lambda_1 - E_1 = \frac{1}{2}(E_1 + E_2) + \frac{1}{2}\epsilon - E_1 = \frac{1}{2}(E_2 - E_1) + \frac{\epsilon}{2} \\ \lambda_2 - E_1 = \Delta - \frac{\epsilon}{2}, \quad \lambda_2 - \lambda_1 = -\epsilon \\ \omega^2 + (\lambda_1 - E_1)^2 = \omega^2 + (\Delta + \frac{\epsilon}{2})^2 \quad \left| \quad \lambda_1 = \overbrace{E_1 + \Delta}^{\beta} + \frac{\epsilon}{2} \right. \\ \omega^2 + (\lambda_2 - E_1)^2 = \omega^2 + (\Delta - \frac{\epsilon}{2})^2 \quad \left| \quad \lambda_2 = E_1 + \Delta - \frac{\epsilon}{2} \right. \end{array} \right.$$

$$= \omega \left\{ \begin{array}{l} \frac{\Delta + \frac{\epsilon}{2}}{\omega^2 + (\Delta + \frac{\epsilon}{2})^2} e^{-\frac{i}{\hbar} t_1 \beta} e^{-\frac{i}{\hbar} t_1 \frac{\epsilon}{2}} + \\ + \frac{\Delta - \frac{\epsilon}{2}}{\omega^2 + (\Delta - \frac{\epsilon}{2})^2} e^{-\frac{i}{\hbar} t_1 \beta} e^{+\frac{i}{\hbar} t_1 \frac{\epsilon}{2}} \end{array} \right\} =$$

$$= \omega e^{-\frac{i}{\hbar} t_1 \beta} \left\{ \left[\frac{\Delta + \frac{\hbar}{2}}{\omega^2 + (\Delta + \frac{\hbar}{2})^2} \right] e^{-i\gamma} \right. +$$

$$\left. \left[\frac{\Delta - \frac{\hbar}{2}}{\omega^2 + (\Delta - \frac{\hbar}{2})^2} \right] e^{i\gamma} \right\}, \quad \gamma = \frac{t_1}{\hbar} \frac{\hbar}{2}$$

B

$$= \omega e^{-\frac{i}{\hbar} t_1 \beta} \left\{ A \cos \gamma - i A \sin \gamma + B \cos \gamma + i B \sin \gamma \right\} =$$

$$= \omega e^{-\frac{i}{\hbar} t_1 \beta} \left\{ (A+B) \cos \gamma - i (A-B) \sin \gamma \right\}$$

$$A+B = \frac{(\Delta + \frac{\hbar}{2}) [\omega^2 + (\Delta - \frac{\hbar}{2})^2] + (\Delta - \frac{\hbar}{2}) [\omega^2 + (\Delta + \frac{\hbar}{2})^2]}{\left\{ \left[\omega^2 + (\Delta + \frac{\hbar}{2})^2 \right] \left[\omega^2 + (\Delta - \frac{\hbar}{2})^2 \right] \right\}}$$

numerator:

$$\left(\Delta + \frac{\hbar}{2}\right) \omega^2 + \left(\Delta + \frac{\hbar}{2}\right) \left(\Delta - \frac{\hbar}{2}\right)^2 + \left(\Delta - \frac{\hbar}{2}\right) \omega^2 + \left(\Delta - \frac{\hbar}{2}\right) \left(\Delta + \frac{\hbar}{2}\right)^2 =$$

$$\left(\Delta + \frac{z}{2}\right) \omega^2 + \left(\Delta + \frac{z}{2}\right) \left(\Delta - \frac{z}{2}\right)^2 + \left(\Delta - \frac{z}{2}\right) \omega^2 + \left(\Delta - \frac{z}{2}\right) \left(\Delta + \frac{z}{2}\right)^2 =$$

$$= \Delta \omega^2 + \cancel{\frac{z}{2} \omega^2} + \Delta \omega^2 - \cancel{\frac{z}{2} \omega^2} +$$

$$+ \left(\Delta + \frac{z}{2}\right) \left(\Delta^2 + \frac{z^2}{4} - z\Delta\right) +$$

$$+ \left(\Delta - \frac{z}{2}\right) \left(\Delta^2 + \frac{z^2}{4} + z\Delta\right) =$$

$$= 2 \cdot \Delta \cdot \omega^2 + 2\Delta \left(\Delta^2 + \frac{z^2}{4}\right) - \cancel{z\Delta^2} + \frac{z}{2} \left(\Delta^2 + \frac{z^2}{4}\right) - \frac{z^2}{2} \Delta +$$

$$+ \cancel{z\Delta^2} - \frac{z}{2} \left(\Delta^2 + \frac{z^2}{4}\right) - \frac{z^2}{2} \Delta =$$

$$= 2\Delta \omega^2 + 2\Delta^3 + \frac{z^2}{2} \Delta - \frac{z^2}{2} \Delta - \frac{z^2}{2} \Delta =$$

$$= 2\Delta \omega^2 + 2\Delta^3 - \frac{z^2}{2} \Delta =$$

$$= 2\Delta \left(\Delta^2 + \omega^2 - \frac{z^2}{4}\right) = 2\Delta \left(\frac{1}{4}(E_2 - E_1)^2 + \omega^2 - \frac{1}{4}(E_2 - E_1)^2 - \omega^2\right) = 0$$

$$\Delta = \frac{1}{2}(E_2 - E_1)$$

$$z^2 = (E_2 - E_1)^2 + 4\omega^2$$

$$\Rightarrow \langle \psi_2 | \psi'(t_1) \rangle = -i\omega(A-B) e^{-\frac{i}{\hbar} t_1 \beta} \text{ sen } \gamma$$

$$\langle \psi_2 | \psi'(t_1) \rangle = -i\omega(A-B) e^{-\frac{i}{\hbar} t_1 \beta} \sin \gamma$$

$$\langle \psi_2 | \psi'(t_1) \rangle = 0 \Rightarrow |\psi'(t_1)\rangle \equiv |\psi_1\rangle$$

$$\sin \gamma = 0$$

$$\sin \frac{t_1}{\hbar} \frac{\hbar}{2} = 0, \quad \frac{t_1}{\hbar} \frac{\hbar}{2} = n\pi, \quad n=0,1,2,\dots$$

$$t_1 = \frac{2n\pi\hbar}{\hbar}, \quad n=0,1,2,\dots \quad \checkmark$$

$$c) P(|\psi'(t)\rangle = |\psi_2\rangle) = |\langle \psi_2 | \psi'(t) \rangle|^2 =$$

$$= \omega^2(A-B)^2 \sin^2\left(\frac{\hbar}{2\hbar} t\right) \quad \checkmark$$

{ Exercise 4 }

Two identical fermions (let's consider spin $1/2$ particles)

- move along $x \in [0, L]$ (infinite square well)
- have spin component along z equal to $\hbar/2$

$$a) \Psi(x_1, x_2; s_1, s_2) = \psi(x_1, x_2) \chi(s_1, s_2)$$

$$\chi(s_1, s_2) = \chi\left(\begin{matrix} + \\ \uparrow \end{matrix}, \begin{matrix} + \\ \uparrow \end{matrix}\right) \rightarrow \text{symmetric under exchange of particles}$$

$\Rightarrow \psi(x_1, x_2)$ must be antisymmetric

So, the wavefunction with minimum energy is:

$$\psi(x_1, x_2) = \frac{1}{\sqrt{2}} \left(\psi_1(x_1) \psi_2(x_2) - \psi_1(x_2) \psi_2(x_1) \right)$$

$$\text{where } \psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \rightarrow \text{eigenfunctions for } H \text{ of the infinite square well}$$

Thus, the minimum energy level

is:

$$E_{n_1=1, n_2=2} = \frac{\hbar^2 \pi^2}{2mL^2} \left(\overset{n_1^2}{1} + \overset{n_2^2}{4} \right)$$

if $n_1 = n_2 = 1$, the wavefunction would vanish.

$$\Rightarrow \psi(x_1, x_2; s_1, s_2) = \left[\frac{1}{\sqrt{2}} \left(\psi_1(x_1) \psi_2(x_2) - \psi_1(x_2) \psi_2(x_1) \right) \right] \chi(+, +)$$

$\downarrow$
 $\psi(x_1, x_2)$

$$E = \frac{5 \pi^2 \hbar^2}{2mL^2}$$

$$b) E^{(1)} = \langle \psi^{(0)} | H' | \psi^{(0)} \rangle =$$

$$= \overbrace{\chi^\dagger(+, +) \chi(+, +)}^1.$$

$$= k \int dx_1 \int dx_2 \psi^*(x_1, x_2) \delta(x_1 - x_2) \psi(x_1, x_2) =$$

$$= k \int dx_1 |\psi(x_1, x_1)|^2 = 0, \text{ since } \psi(x, x) \equiv 0 \text{ (antisymmetric state)}$$

c) if $\chi(s_1, s_2) = \chi(s=0, m_s=0)$ (spin singlet)
 - antisymmetric -
 Then the spatial wavefunction corresponding to the minimum energy is:

$$\psi(x_1, x_2) = \psi_1(x_1) \psi_1(x_2) \rightarrow \text{symmetric} \quad \text{must be}$$

$$E = \frac{\pi^2 \hbar^2}{mL^2}$$

$$E^{(1)} = \langle \psi^{(0)} | H' | \psi^{(0)} \rangle =$$

$$= \overbrace{\chi^\dagger(0,0) \chi(0,0)}^1.$$

$$k \int dx_1 \int dx_2 \psi^*(x_1, x_2) \delta(x_1 - x_2) \psi(x_1, x_2) =$$

$$= k \int dx_1 |\psi(x_1, x_1)|^2 = k \int_0^L dx |\psi_1(x)|^4 =$$

$$= k \frac{k}{L^2} \frac{3}{8} k = \frac{3k}{2L}$$

↙ ✓

calculation of the integral

$$\int_0^L dx \sin^4 \frac{\pi x}{L} = \int_0^L dx \sin^2 \frac{\pi x}{L} \left(1 - \cos^2 \frac{\pi x}{L} \right) =$$

$$= \frac{L}{2} - \int_0^L dx \sin^2 \frac{\pi x}{L} \cos^2 \frac{\pi x}{L} =$$

$$= \frac{L}{2} - \int_0^L dx \frac{1}{4} \sin^2 \frac{2\pi x}{L} =$$

$$\alpha = \frac{2\pi x}{L}$$

$$d\alpha = \frac{2\pi}{L} dx$$

$$= \frac{L}{2} - \int_0^{2\pi} d\alpha \frac{L}{8\pi} \sin^2 \alpha =$$

$$\cos^2 \alpha = \frac{1}{2} + \frac{1}{2} \cos 2\alpha$$

$$= \frac{L}{2} - \frac{L}{8\pi} \int_0^{2\pi} d\alpha (1 - \cos^2 \alpha) =$$

$$= \frac{L}{2} - \frac{L}{8\pi} \int_0^{2\pi} d\alpha \left(\frac{1}{2} - \frac{1}{2} \cos 2\alpha \right) =$$

$$= \frac{L}{2} - \frac{L}{8\pi} \int_0^{2\pi} d\alpha \left(\frac{1}{2} - \frac{1}{2} \cos 2\alpha \right) =$$

$$= \frac{L}{2} - \frac{L}{8\pi} \frac{1}{2} 2\pi + \frac{L}{16\pi} \int_0^{2\pi} d\alpha \cos 2\alpha =$$

$$= \frac{L}{2} - \frac{L}{8} + \frac{\frac{d}{d\alpha} \sin(2\alpha) = \cos(2\alpha) \cdot 2}{32\pi} \sin(2\alpha) \Big|_0^{2\pi} =$$

$$= \frac{3}{8} L.$$